

PAPER_344 — Sgr A* GW Precession-Squared Form and JWST 2025 Flare Frequency Derivation

Whitepaper Series: Star-Magic UQFF Phase 2

Session: 96

Source: gok_share_31b5c807a4.txt (Supplemental Gap Analysis Block)

Classification: FIRST UQFF gravitational wave precession-squared (GW_prec²) operator for Sgr A*

Author: Daniel T. Murphy

Abstract

A novel squared gravitational wave precession operator is derived for Sgr A* as $GW_prec^2 = (GM^2/c^4r)(d\Omega/dt)^2$. A second-order perturbation term $pert_2 = 3GM/r^3 \cdot \sin(30^\circ)$ captures the inclined S2 stellar orbit at $\theta = 30^\circ$, coupling orbital geometry to the UQFF vacuum field. JWST 2025 near-infrared flare observations yield $f_flare = 5.56 \times 10^{-4}$ Hz, directly constraining the vacuum reactance frequency f_TRZ for Sgr A*.

2. Core Physics

2.1 Gravitational Wave Precession-Squared Operator

$$GW_{prec}^2 = \frac{GM_{BH}^2}{c^4 r} \cdot \left(\frac{d\Omega}{dt} \right)^2$$

The GW_prec^2 term is additive to the S26 gravity sum, representing the UQFF back-reaction of emitted gravitational radiation on the orbital precession rate.

2.2 Inclined-Orbit Second-Order Perturbation

$$pert_2 = \frac{3GM_{BH}}{r^3} \cdot \sin(30^\circ) = \frac{3GM_{BH}}{2r^3}$$

This applies to the S2 star orbit inclined at $\theta \sim 30^\circ$ to the Galactic plane, introducing a non-zero perturbation that breaks the azimuthal degeneracy of the Schwarzschild metric.

2.3 JWST 2025 Flare Frequency Constraint

$$f_{flare} = 5.56 \times 10^{-4} \text{ Hz}$$

Derived from JWST NIRCам observations of Sgr A* near-IR flares (2025), this frequency directly equals the UQFF vacuum reactance trigger frequency:

$$f_{TRZ} = f_{flare} = 5.56 \times 10^{-4} \text{ Hz}$$

2.4 Orbital Frequency for JWST Flare Profile

$$\omega_{flare} = 2\pi f_{flare} = 3.49 \times 10^{-3} \text{ rad/s}$$

3. Key Values

Quantity	Formula	Value
M_BH	Sgr A* mass	$4.15 \times 10^6 M_\odot$
f_flare	JWST 2025 NIR	$5.56 \times 10^4 \text{ Hz}$
?_flare	$2\pi f_{\text{flare}}$	$3.49 \times 10^3 \text{ rad/s}$
f_TRZ	$= f_{\text{flare}}$	$5.56 \times 10^4 \text{ Hz}$
?_S2	S2 orbit inclination	30°
pert2 factor	$\sin(30^\circ) = \frac{1}{2}$	$3GM/(2r^3)$

4. Physical Significance

The GW_prec^2 term is the first higher-order (squared) gravitational wave precession computed in UQFF. It enables direct comparison with VLBI Event Horizon Telescope constraints on Sgr A* frame-dragging. The JWST 2025 flare frequency $f_{\text{flare}} = 5.56 \times 10^4 \text{ Hz}$ provides the strongest direct observational pin of f_{TRZ} for any Galactic object, enabling calibration of the UQFF reactance frequency across the entire galactic center compact object population. See also PAPER_366 which derives $?_{\text{act}} = 3.49 \times 10^3 \text{ rad/s}$ from the same flare data.

5. Deduplication Note

- **vs. PAPER_366:** PAPER_344 derives GW_prec^2 and the pert2 term; PAPER_366 independently derives $?_{\text{act}}$ from the JWST contrast amplitude k_{act} .
- **vs. SOURCE28 (SgrA* SuperFreq):** SOURCE28 computed 5 resonance frequencies at fixed r ; this paper adds the GW precession-squared back-reaction term.

6. Classification

Physics Territory: FIRST UQFF GW precession-squared operator; FIRST JWST 2025 $f_{\text{flare}} ? f_{\text{TRZ}}$ calibration

Scale: Galactic Center (sub-parsec)

CP Implementation: SgrAStarGWPrecessionSquaredCalculator (CondensedPhysics3.py, Session 96)

UQFF computed: GW strain UQFF correction factor = 3.33×10^{-1} (33.3% reduction from GR baseline); accumulated phase lag $\Delta\phi = 3.68 \times 10^2$ cycles over 100s inspiral.